

Unbounded metric-weak convergence in the whole Banach space c_0

Candidate counterexample packet for arXiv:2506.04154

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Status: candidate full counterexample, likely valid, subject to human review. In the Banach space c_0 with its supremum norm, the unbounded sequence $x_n = ne_n$ converges d -weakly to 0 in the sense introduced by Gutiérrez and Nevanlinna. More strongly, the amplitudes may be arbitrary, and the same conclusion holds in every sup-norm coordinate space containing the coordinate vectors. This disproves Conjecture 1.5 of the source.

1 Source conjecture

For a pointed metric space $(X, d, 0)$ and $w \in X$, the source defines the internal metric functional

$$h_w(x) = d(x, w) - d(0, w).$$

The metric functionals $X^\diamond$ are the pointwise closure of the internal ones. Thus every $h \in X^\diamond$ is the pointwise limit of a net (h_{w_α}) . A sequence (x_n) converges d -weakly to z when

$$\liminf_{n \rightarrow \infty} h(x_n) \geq h(z) \quad \text{for every } h \in X^\diamond. \quad (1)$$

After proving that unbounded sequences cannot converge d -weakly in ℓ_1 , in $C[0, 1]$, or when the dual norm is strictly convex, Gutiérrez and Nevanlinna state [1, Conjecture 1.5]:

“There is no normed linear space where unbounded sequences converge d -weakly.”

The source pages containing the definition and conjecture are reproduced in Appendix A.

Idea of the proof

An internal functional on a sup-norm space cannot be negative by a fixed amount at two vectors supported on different coordinates. Indeed, negativity at ae_i forces every coordinate of its centre except possibly i to lie strictly below the centre’s supremum norm; negativity at be_j , $j \neq i$, also controls coordinate i . Together they say every coordinate is below the supremum by a uniform amount, a contradiction. Pointwise convergence of a net is simultaneous on any fixed finite set, so the same two-point exclusion holds for every metric functional. Consequently, along a sequence of distinct coordinate spikes, any metric functional can dip below $-\varepsilon$ at most once. This is exactly the liminf condition at zero.

2 The two-spike exclusion

Let I be an infinite set and let Y be a real linear subspace of $\ell_\infty(I)$, equipped with the inherited supremum norm, such that $e_i \in Y$ for every $i \in I$. This includes $c_0(I)$ and $\ell_\infty(I)$. We use basepoint 0.

Lemma 1 (internal two-spike exclusion). *Let $w \in Y$, let $i \neq j$, let $a, b \in \mathbb{R}$, and let $\varepsilon > 0$. It is impossible that both*

$$h_w(ae_i) < -\varepsilon \quad \text{and} \quad h_w(be_j) < -\varepsilon.$$

Proof. Put $r = \|w\|_\infty$. The first inequality says

$$\|ae_i - w\|_\infty < r - \varepsilon.$$

In particular, $|w_k| < r - \varepsilon$ for every $k \neq i$. The second inequality similarly gives $|w_k| < r - \varepsilon$ for every $k \neq j$. Because $i \neq j$, the two collections cover all $k \in I$. Hence

$$\|w\|_\infty = \sup_{k \in I} |w_k| \leq r - \varepsilon < r,$$

a contradiction. (If $r - \varepsilon \leq 0$, either original inequality is already impossible.) □

Lemma 2 (closure preserves exclusion). *For every metric functional $h \in Y^\diamond$, every $i \neq j$, every $a, b \in \mathbb{R}$, and every $\varepsilon > 0$, it is impossible that both*

$$h(ae_i) < -\varepsilon \quad \text{and} \quad h(be_j) < -\varepsilon.$$

Proof. Suppose both inequalities hold. Choose a net of internal functionals h_{w_α} converging pointwise to h . Pointwise convergence on the two-element set $\{ae_i, be_j\}$ supplies a common tail on which

$$h_{w_\alpha}(ae_i) < -\varepsilon/2, \quad h_{w_\alpha}(be_j) < -\varepsilon/2.$$

This contradicts Lemma 1. □

3 The counterexample

Theorem 1 (arbitrary coordinate amplitudes). *Let (i_n) be pairwise distinct elements of I and let (a_n) be any real sequence. In the whole normed linear space Y ,*

$$a_n e_{i_n} \xrightarrow{d} 0.$$

In particular, $x_n = ne_n$ is an unbounded d -weakly convergent sequence in c_0 .

Proof. Every internal functional vanishes at the basepoint, so every metric functional does as well: $h(0) = 0$ for all $h \in Y^\diamond$. Fix such an h and $\varepsilon > 0$. Lemma 2 shows that at most one index n can satisfy

$$h(a_n e_{i_n}) < -\varepsilon.$$

Therefore, for every $\varepsilon > 0$, the sequence is eventually bounded below by $-\varepsilon$. It follows that

$$\liminf_{n \rightarrow \infty} h(a_n e_{i_n}) \geq 0 = h(0).$$

Since this holds for every metric functional, definition (1) gives d -weak convergence to zero. Taking $Y = c_0$, $i_n = n$, and $a_n = n$ gives $\|x_n\|_\infty = n$, so the sequence is unbounded. □

Corollary 1. *Conjecture 1.5 of arXiv:2506.04154 is false. It already fails in the classical separable Banach space c_0 .*

4 Checks, scope, and novelty

The proof uses only the definitions and the elementary supremum-norm argument in Lemma 1. Its delicate point is the passage from internals to their pointwise closure: a convergent net is eventually close at both fixed test points simultaneously, which is precisely what Lemma 2 records. No sequential compactness or explicit classification of the metric compactification of c_0 is assumed.

The source itself observes that the non-linear metric subspace $\{0, e_1, 2e_2, 3e_3, \dots\} \subset \ell_1$ admits unbounded d -weak convergence, while its Theorem 1.4 rules this out in the *whole* space ℓ_1 . Theorem 1 supplies exactly the missing ambient-linear example. It does not classify all unbounded d -weakly convergent sequences, all Banach spaces admitting them, or their possible limits.

The bounded novelty check on 11 August 2026 searched the run registry and the solution, attempt, and proof-gap indexes for the arXiv id and the central c_0 /coordinate-spike phrases. Exact-title, exact-conjecture, and close-phrase primary-source web searches found the source preprint and unrelated general horofunction literature, but no later answer or matching counterexample. This is not an exhaustive priority claim.

Human-review recommendation. Check the quantifiers in Lemma 2, especially simultaneous approximation at two points by one tail of the net. Once that standard finite-intersection step is accepted, the counterexample is decisive.

A Source-page evidence

References

- [1] Armando W. Gutiérrez and Olavi Nevanlinna, *Metric functionals and weak convergence*, arXiv:2506.04154v1, 2025.

convergence that agrees with the standard one in Hilbert spaces. See the works [KP08], [EFL09], [Bač23], and [LP23].

In this note, we present a novel notion of weak convergence in metric spaces. Our approach uses metric functionals.

Definition 1.1. Let (X, d) be a metric space. We say that a sequence of points $(x_n)_{n \geq 1}$ in X converges *d-weakly* to a point z in X if we have

$$\liminf_{n \rightarrow \infty} \mathbf{h}(x_n) \geq \mathbf{h}(z), \quad (1.1)$$

for all metric functionals $\mathbf{h}$ on X . We use the notation $x_n \xrightarrow{d} z$.

Our concept of *d-weak* convergence can be applied to nets. We discuss here *d-weak* convergence of sequences only to simplify the exposition of our ideas.

We also want to emphasize that *d-weak* convergence fully depends on the given metric d . The metric d determines all the corresponding metric functionals $\mathbf{h}$ that are essential to test the inequality (1.1).

The concept of a metric functional is fairly known in some mathematical disciplines. Nonetheless, we recall such a concept and provide new properties in Section 2.

We proceed now to provide evidence that supports our choice of Definition 1.1 as a suitable candidate for weak convergence in metric spaces. The following nontrivial statements concern *d-weak* convergence of sequences in normed linear spaces.

Theorem 1.2. *Let $(X, \|\cdot\|)$ be a normed linear space. Let d be the metric on X induced by the norm $\|\cdot\|$. If a sequence of points $(x_n)_{n \geq 1}$ in X converges *d-weakly*, then its limit is unique.*

The following result shows the connection of *d-weak* convergence with the standard weak convergence in normed linear spaces. Namely, they agree for bounded sequences in *all* normed linear spaces.

Theorem 1.3. *Let $(X, \|\cdot\|)$ be a normed linear space. Let d be the metric on X induced by the norm $\|\cdot\|$. Suppose that $(x_n)_{n \geq 1}$ is a bounded sequence in X and z is a point in X . Then, $(x_n)_{n \geq 1}$ converges *d-weakly* to z if and only if $(x_n)_{n \geq 1}$ converges weakly to z .*

One may wonder whether unbounded sequences in normed linear spaces converge *d-weakly*. We can prove that this does not happen in some normed linear spaces.

Figure 1: Page 2 of arXiv:2506.04154v1, containing Definition 1.1 of *d-weak* convergence and the start of the normed-space results.

Theorem 1.4. *Suppose that X is ℓ_1 , or $C[0,1]$, or a normed linear space whose dual is strictly convex. If d is the corresponding induced metric, then unbounded sequences in X do not converge d -weakly.*

Conjecture 1.5. *There is no normed linear space where unbounded sequences converge d -weakly.*

Without a proof of the above conjecture, it is unclear how d -weak convergence behaves on linear combinations of two d -weakly convergent sequences. Nevertheless, we can offer the following.

Theorem 1.6. *Let $(X, \|\cdot\|)$ be a normed linear space. Let d be the metric on X induced by the norm $\|\cdot\|$. Suppose that $(x_n)_{n \geq 1}$ and $(y_n)_{n \geq 1}$ are two sequences in X . If there are two vectors u and v in X such that $x_n \xrightarrow{d} u$ and $d(y_n, v) \rightarrow 0$, then we have*

$$sx_n + ty_n \xrightarrow{d} su + tv,$$

for all real numbers s and t .

All the results presented above examine d -weak convergence of sequences in the whole normed linear space. Proper subsets become legit metric spaces when we equip them with the subspace topology. With this in mind, we can produce simple examples of metric spaces where unbounded sequences converge d -weakly, or where d -weakly convergent sequences have more than one limit. An example where both situations happen is the following. Equip the set $X = \{0, e_1, 2e_2, 3e_3, \dots\}$ with the metric d that is induced by the norm of ℓ_1 . We can verify that the unbounded sequence $(ne_n)_{n \geq 1}$ in X converges d -weakly to every point in X . More examples are presented in Section 3.

The following result states that the set of all d -weak limits of a sequence in a W -convex metric space is also W -convex. The concept of W -convexity was introduced by W. Takahashi in [Tak70]. We recall this concept in Definition 2.7. Briefly we mention that every metric space that admits a conical geodesic bicombing is W -convex.

Theorem 1.7. *Let (X, d) be a W -convex metric space. Then, the set of all d -weak limits of a sequence in X is W -convex and closed.*

2 Metric functionals

2.1 Where do they appear?

Metric functionals have become valuable tools in investigations of iterative processes in metric structures.

Figure 2: Page 3 of arXiv:2506.04154v1. Conjecture 1.5 rules out unbounded d -weak convergence in any normed linear space; the same page distinguishes this from examples on proper metric subsets.